

svd_bb

August 6, 2024

1 Initial PCA Work Using SVD

```
[ ]: # Data analysis
import pandas as pd
import numpy as np

# Plotting and Correlation Maths
import seaborn as sns
import scipy as sci

# Simple model development
import sklearn as sk
from sklearn.preprocessing import MinMaxScaler, StandardScaler

import matplotlib.pyplot as plt

import plotly as px

import os
```

2 Read Data

```
[ ]: # get data
data = pd.read_csv('../data/processed/cleaned_data.csv')
```

```
[ ]: data
```

```
[ ]:      tempmax  tempmin  temp  feelslikemax  feelslikemin  feelslike  dew  \
0         26.2      -4.2   10.1           18.1          -16.6         0.6  -3.8
1         35.8      28.0   32.5           34.7           24.1        26.8  26.6
2         15.3       2.5    9.2           14.6           -6.0         3.5   0.5
3         15.3       2.5    9.2           14.6           -6.0         3.5   0.5
4         15.3       2.5    9.2           14.6           -6.0         3.5   0.5
...      ...      ...   ...           ...           ...      ...
3679      68.8      58.9   61.6           68.8          58.9        61.6  56.7
3680      59.4      46.8   50.8           59.4          39.8        46.5  46.3
```

3681	59.2	44.5	51.1	59.2	37.4	47.9	33.0
3682	81.1	45.0	64.1	79.9	39.9	62.9	39.5
3683	81.1	45.0	64.1	79.9	39.9	62.9	39.5

	humidity	precip	precipcover	...	preciptype_freezingrain	\
0	57.2	0.000	0.00	...	0	
1	78.9	0.000	0.00	...	0	
2	68.0	0.000	0.00	...	0	
3	68.0	0.000	0.00	...	0	
4	68.0	0.000	0.00	...	0	
...	...	...	...	...	...	
3679	84.8	1.526	83.33	...	0	
3680	84.5	0.266	83.33	...	0	
3681	51.5	0.019	12.50	...	0	
3682	42.6	0.001	4.17	...	0	
3683	42.6	0.001	4.17	...	0	

	preciptype_freezingrain,snow	preciptype_freezingrain,snow,ice	\
0	0	0	
1	0	0	
2	0	0	
3	0	0	
4	0	0	
...	...	...	
3679	0	0	
3680	0	0	
3681	0	0	
3682	0	0	
3683	0	0	

	preciptype_none	preciptype_rain	preciptype_rain,freezingrain	\
0	1	0	0	
1	1	0	0	
2	1	0	0	
3	1	0	0	
4	1	0	0	
...	...	...	...	
3679	0	1	0	
3680	0	1	0	
3681	0	1	0	
3682	0	1	0	
3683	0	1	0	

	preciptype_rain,freezingrain,snow	preciptype_rain,snow	\
0	0	0	
1	0	0	
2	0	0	

```

3          0          0
4          0          0
...
3679      0          0
3680      0          0
3681      0          0
3682      0          0
3683      0          0

```

```

preciptype_rain,snow,ice  preciptype_snow
0          0          0
1          0          0
2          0          0
3          0          0
4          0          0
...
3679      0          0
3680      0          0
3681      0          0
3682      0          0
3683      0          0

```

[3684 rows x 26 columns]

3 Scale the Data for SVD

```
[ ]: scaler = StandardScaler()
```

```
scaled_data = scaler.fit_transform(data)
```

```
[ ]: scaled_data_df = pd.DataFrame(scaled_data, columns=data.columns)
scaled_data_df
```

```

[ ]:      tempmax  tempmin    temp  feelslikemax  feelslikemin  feelslike  \
0    -2.072065 -3.155709 -2.681172    -2.200051    -3.053488 -2.664663
1    -1.533370 -1.159988 -1.333912    -1.372703    -0.981127 -1.320704
2    -2.683709 -2.740451 -2.735303    -2.374492    -2.513758 -2.515904
3    -2.683709 -2.740451 -2.735303    -2.374492    -2.513758 -2.515904
4    -2.683709 -2.740451 -2.735303    -2.374492    -2.513758 -2.515904
...
3679  0.318396  0.755160  0.416323     0.326850     0.790818  0.464403
3680 -0.209077  0.005215 -0.233248    -0.141648    -0.181715 -0.310169
3681 -0.220300 -0.137336 -0.215205    -0.151616    -0.303918 -0.238355
3682  1.008599 -0.106347  0.566687     0.880077    -0.176623  0.531088
3683  1.008599 -0.106347  0.566687     0.880077    -0.176623  0.531088

      dew  humidity  precip  precipcover  ...  preciptype_freezingrain  \

```

0	-2.760187	-0.871929	-0.431235	-0.714985	...	-0.049487
1	-0.991367	0.695315	-0.431235	-0.714985	...	-0.049487
2	-2.509992	-0.091918	-0.431235	-0.714985	...	-0.049487
3	-2.509992	-0.091918	-0.431235	-0.714985	...	-0.049487
4	-2.509992	-0.091918	-0.431235	-0.714985	...	-0.049487
...	...	...	...	...	...	...
3679	0.759998	1.121433	4.714725	3.110540	...	-0.049487
3680	0.154875	1.099766	0.465768	3.110540	...	-0.049487
3681	-0.618983	-1.283602	-0.367163	-0.141133	...	-0.049487
3682	-0.240782	-1.926389	-0.427862	-0.523548	...	-0.049487
3683	-0.240782	-1.926389	-0.427862	-0.523548	...	-0.049487

	preciptype_freezingrain,snow	preciptype_freezingrain,snow,ice	\
0	-0.052171	-0.032969	
1	-0.052171	-0.032969	
2	-0.052171	-0.032969	
3	-0.052171	-0.032969	
4	-0.052171	-0.032969	
...	...	...	
3679	-0.052171	-0.032969	
3680	-0.052171	-0.032969	
3681	-0.052171	-0.032969	
3682	-0.052171	-0.032969	
3683	-0.052171	-0.032969	

	preciptype_none	preciptype_rain	preciptype_rain,freezingrain	\
0	1.297373	-0.956952	-0.057166	
1	1.297373	-0.956952	-0.057166	
2	1.297373	-0.956952	-0.057166	
3	1.297373	-0.956952	-0.057166	
4	1.297373	-0.956952	-0.057166	
...	...	...	...	
3679	-0.770789	1.044985	-0.057166	
3680	-0.770789	1.044985	-0.057166	
3681	-0.770789	1.044985	-0.057166	
3682	-0.770789	1.044985	-0.057166	
3683	-0.770789	1.044985	-0.057166	

	preciptype_rain,freezingrain,snow	preciptype_rain,snow	\
0	-0.032969	-0.338637	
1	-0.032969	-0.338637	
2	-0.032969	-0.338637	
3	-0.032969	-0.338637	
4	-0.032969	-0.338637	
...	...	...	
3679	-0.032969	-0.338637	
3680	-0.032969	-0.338637	

3681	-0.032969	-0.338637
3682	-0.032969	-0.338637
3683	-0.032969	-0.338637

	preciptype_rain,snow,ice	preciptype_snow
0	-0.04039	-0.188184
1	-0.04039	-0.188184
2	-0.04039	-0.188184
3	-0.04039	-0.188184
4	-0.04039	-0.188184
...	...	...
3679	-0.04039	-0.188184
3680	-0.04039	-0.188184
3681	-0.04039	-0.188184
3682	-0.04039	-0.188184
3683	-0.04039	-0.188184

[3684 rows x 26 columns]

```
[ ]: scaled_data_df.describe()
```

```
[ ]:
```

	tempmax	tempmin	temp	feelslikemax	feelslikemin	\
count	3.684000e+03	3.684000e+03	3.684000e+03	3.684000e+03	3.684000e+03	
mean	-4.088900e-16	3.471707e-16	9.257886e-17	-1.311534e-16	-1.234385e-16	
std	1.000136e+00	1.000136e+00	1.000136e+00	1.000136e+00	1.000136e+00	
min	-2.902554e+00	-3.205292e+00	-3.060089e+00	-2.967591e+00	-3.756156e+00	
25%	-8.207206e-01	-8.624894e-01	-8.527477e-01	-8.493800e-01	-8.741995e-01	
50%	2.061676e-01	4.860062e-02	1.035666e-01	2.271696e-01	1.085172e-01	
75%	8.683140e-01	9.240528e-01	9.035024e-01	8.152846e-01	9.295693e-01	
max	1.715637e+00	1.784010e+00	1.721482e+00	1.911770e+00	1.636056e+00	

	feelslike	dew	humidity	precip	precipcover	\
count	3.684000e+03	3.684000e+03	3.684000e+03	3.684000e+03	3.684000e+03	
mean	2.391621e-16	-1.542981e-16	3.857452e-16	-6.171924e-17	-1.542981e-17	
std	1.000136e+00	1.000136e+00	1.000136e+00	1.000136e+00	1.000136e+00	
min	-3.372550e+00	-2.952197e+00	-3.125295e+00	-4.312346e-01	-7.149849e-01	
25%	-8.641677e-01	-8.459044e-01	-6.642873e-01	-4.312346e-01	-7.149849e-01	
50%	1.771444e-01	3.268714e-02	8.502896e-02	-4.143737e-01	-5.235480e-01	
75%	8.799019e-01	9.112787e-01	7.530941e-01	-5.860779e-02	4.327184e-01	
max	1.777585e+00	1.795689e+00	2.161447e+00	1.279449e+01	3.875828e+00	

	...	preciptype_freezingrain	preciptype_freezingrain,snow	\
count	...	3.684000e+03	3.684000e+03	
mean	...	-1.157236e-17	-1.542981e-17	
std	...	1.000136e+00	1.000136e+00	
min	...	-4.948717e-02	-5.217115e-02	
25%	...	-4.948717e-02	-5.217115e-02	

50%	...	-4.948717e-02	-5.217115e-02
75%	...	-4.948717e-02	-5.217115e-02
max	...	2.020726e+01	1.916768e+01

	preciptype_freezingrain,snow,ice	preciptype_none	preciptype_rain \
count	3.684000e+03	3.684000e+03	3.684000e+03
mean	-1.157236e-17	-7.714905e-17	4.628943e-17
std	1.000136e+00	1.000136e+00	1.000136e+00
min	-3.296902e-02	-7.707886e-01	-9.569517e-01
25%	-3.296902e-02	-7.707886e-01	-9.569517e-01
50%	-3.296902e-02	-7.707886e-01	-9.569517e-01
75%	-3.296902e-02	1.297373e+00	1.044985e+00
max	3.033150e+01	1.297373e+00	1.044985e+00

	preciptype_rain,freezingrain	preciptype_rain,freezingrain,snow \
count	3.684000e+03	3.684000e+03
mean	-4.628943e-17	-1.350108e-17
std	1.000136e+00	1.000136e+00
min	-5.716620e-02	-3.296902e-02
25%	-5.716620e-02	-3.296902e-02
50%	-5.716620e-02	-3.296902e-02
75%	-5.716620e-02	-3.296902e-02
max	1.749286e+01	3.033150e+01

	preciptype_rain,snow	preciptype_rain,snow,ice	preciptype_snow
count	3.684000e+03	3.684000e+03	3.684000e+03
mean	-3.085962e-17	3.857452e-18	-3.085962e-17
std	1.000136e+00	1.000136e+00	1.000136e+00
min	-3.386366e-01	-4.038962e-02	-1.881838e-01
25%	-3.386366e-01	-4.038962e-02	-1.881838e-01
50%	-3.386366e-01	-4.038962e-02	-1.881838e-01
75%	-3.386366e-01	-4.038962e-02	-1.881838e-01
max	2.953018e+00	2.475884e+01	5.313953e+00

[8 rows x 26 columns]

4 Perform SVD

```
[ ]: U, S, V = sci.linalg.svd(scaled_data)
```

```
[ ]: S
```

```
[ ]: array([1.72914301e+02, 1.19750612e+02, 7.14021186e+01, 6.82792702e+01,
        6.48266810e+01, 6.19597871e+01, 6.11608737e+01, 6.09943487e+01,
        6.08499106e+01, 6.07404967e+01, 5.94783658e+01, 5.75535635e+01,
        5.32134990e+01, 4.79005818e+01, 4.52263046e+01, 4.07143322e+01,
        3.23696250e+01, 3.07796613e+01, 2.93955650e+01, 1.65426231e+01,
```

```
5.74898987e+00, 4.98363320e+00, 3.56040349e+00, 2.30321170e+00,  
1.18525850e+00, 4.45168621e-14])
```

5 Plot Singular Values

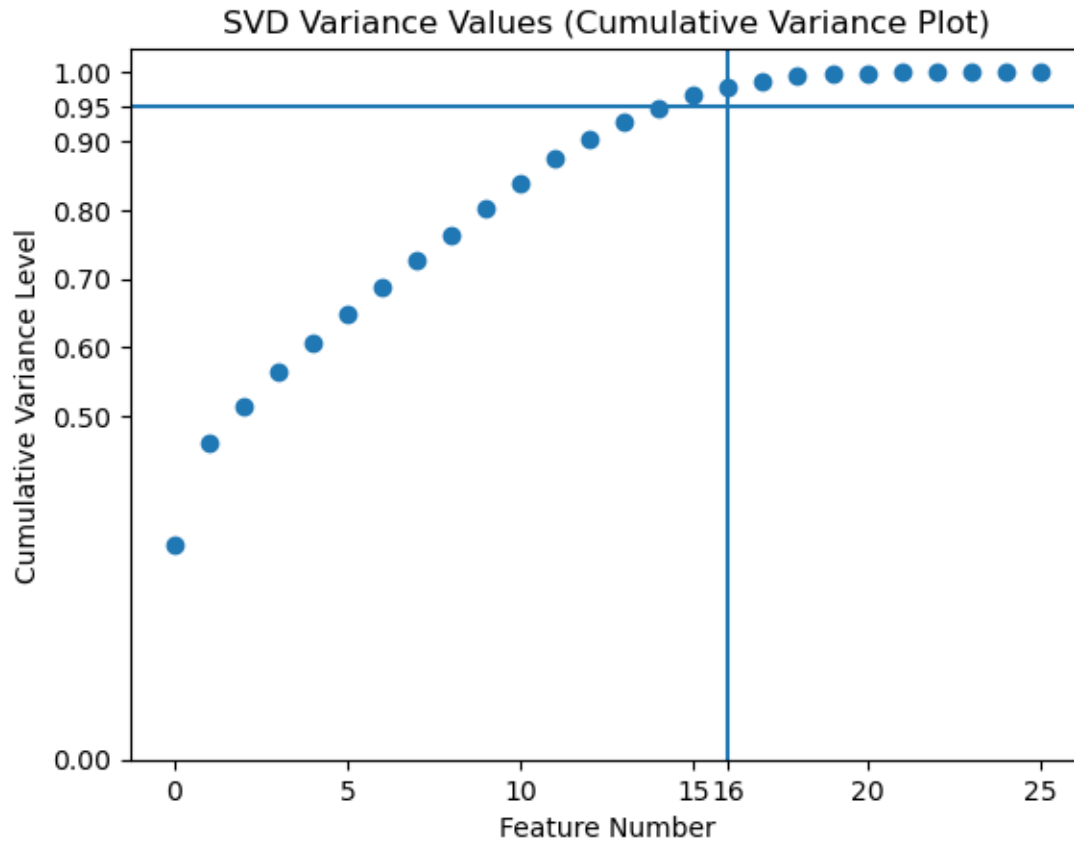
Use these to determine variance cutoff

```
[ ]: s_values = np.cumsum(S**2) / sum(S**2)  
      print(s_values)
```

```
[0.31215397 0.461868  0.51509466 0.56376729 0.60764203 0.64772195  
0.68677495 0.72561558 0.76427247 0.80279047 0.83972436 0.87430647  
0.90386962 0.9278242  0.94917869 0.96648489 0.97742401 0.98731488  
0.99633622 0.99919325 0.99953831 0.99979761 0.99992995 0.99998533  
1.          1.          ]
```

```
[ ]: # Create plot  
fig, ax = plt.subplots()  
plt.scatter(range(0,len(s_values)),s_values)  
plt.axhline(0.95)  
plt.axvline(16)  
plt.xlabel('Feature Number')  
plt.ylabel('Cumulative Variance Level')  
plt.title('SVD Variance Values (Cumulative Variance Plot)')  
  
xt = [0, 5, 10, 15, 16, 20, 25]  
yt = [0, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 1.0]  
plt.xticks(ticks=xt)  
plt.yticks(ticks=yt)  
  
plt.show
```

```
[ ]: <function matplotlib.pyplot.show(close=None, block=None)>
```



Variance cutoff can be done around 15-17 features. The plot shows the intersection at 95% variance and 16 S values.

This will allow us to drop around 10 features and still have reasonable outputs from the model.

6 A BETTER METHOD

```
[ ]: from sklearn.decomposition import TruncatedSVD, PCA
```

```
[ ]: pca = PCA(n_components=26, svd_solver='full')
pca.fit(scaled_data)

out = pca.transform(scaled_data)
```

```
[ ]: sum(pca.explained_variance_ratio_)
```

```
[ ]: 0.9999999999999997
```

```
[ ]: ratios = pca.explained_variance_ratio_
ratios
```

```
[ ]: array([3.12153966e-01, 1.49714034e-01, 5.32266615e-02, 4.86726253e-02,
          4.38747450e-02, 4.00799217e-02, 3.90529992e-02, 3.88406266e-02,
          3.86568907e-02, 3.85179982e-02, 3.69338929e-02, 3.45821084e-02,
          2.95631471e-02, 2.39545825e-02, 2.13544917e-02, 1.73061978e-02,
          1.09391195e-02, 9.89087475e-03, 9.02133178e-03, 2.85703644e-03,
          3.45056424e-04, 2.59298002e-04, 1.32344369e-04, 5.53827795e-05,
          1.46667263e-05, 1.28314364e-32])
```

```
[ ]: r_values = np.cumsum(ratios) / sum(ratios)
      print(np.cumsum(ratios)/sum(ratios))
```

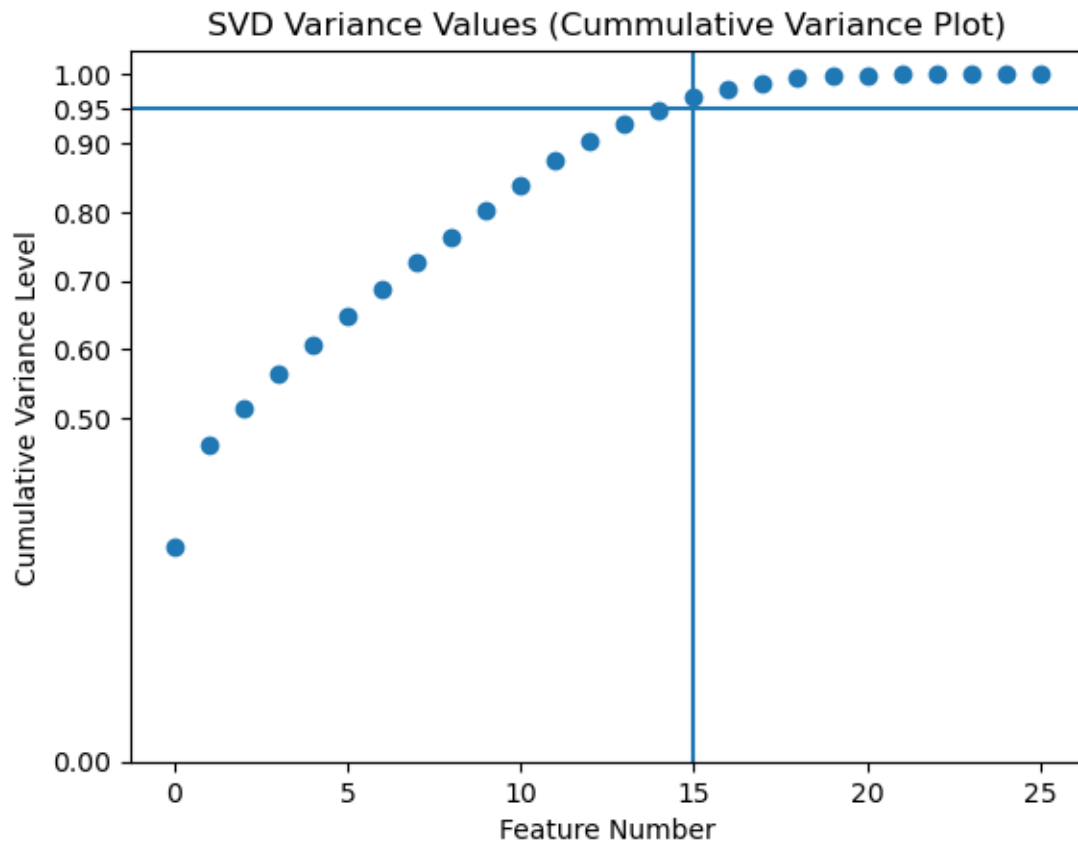
```
[0.31215397 0.461868  0.51509466 0.56376729 0.60764203 0.64772195
 0.68677495 0.72561558 0.76427247 0.80279047 0.83972436 0.87430647
 0.90386962 0.9278242  0.94917869 0.96648489 0.97742401 0.98731488
 0.99633622 0.99919325 0.99953831 0.99979761 0.99992995 0.99998533
 1.          1.          ]
```

```
[ ]: # Create plot
      fig, ax = plt.subplots()
      plt.scatter(range(0,len(r_values)),r_values)
      plt.axhline(0.95)
      plt.axvline(15)
      plt.xlabel('Feature Number')
      plt.ylabel('Cumulative Variance Level')
      plt.title('SVD Variance Values (Cumulative Variance Plot)')

      xt = [0, 5, 10, 15, 20, 25]
      yt = [0, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 1.0]
      plt.xticks(ticks=xt)
      plt.yticks(ticks=yt)

      plt.show
```

```
[ ]: <function matplotlib.pyplot.show(close=None, block=None)>
```



[]: